

Suman Mandava

Buffalo, NY | +1 716-319-4581 | sumanm2003@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

Summary

Software Engineer with experience building AI-powered applications and full-stack products. Hands-on with LLM fine-tuning (LoRA), prompt engineering, and end-to-end model training/evaluation/inference pipelines using PyTorch. Shipped customer-facing systems with measurable improvements in reliability, scalability, and performance/latency, and deployed cloud services using Docker. Excited to apply AI-driven workflows to personalized customer messaging journeys at scale and contribute through code reviews, documentation, and strong engineering fundamentals.

Education

University at Buffalo, The State University of New York

Aug 2024 - Dec 2025

Master of Science, Computer Science and Engineering

NY, USA

- **GPA:** 3.8
- **Coursework:** Data Structures & Algorithms (dynamic programming), Machine Learning, Deep Learning, Database Systems (SQL), Operating Systems, Computer Security

St. Joseph's Institute of Technology

Mar 2020 - Apr 2024

bachelor of engineering, computer science

Chennai, India

- **GPA:** 3.6
- **Coursework:** Data Structures & Algorithms, Operating Systems, Computer Networks, Microprocessors & Microcontroller, Database Management Systems (SQL)

Experience

Kornia (Open Source) - Contributor

Jan 2026 - Present

Deep Learning (PyTorch)

Remote

- Implemented core components of the Qwen2.5-VL vision-language encoder in Kornia, enabling efficient downstream inference workflows for AI-powered applications.
- Built modular PyTorch components optimized for high-dimensional tensors (B, N, 1280), improving performance, readability, and maintainability for scalable AI workflows.
- Strengthened release readiness by resolving CI checks (Ruff, Mypy) and expanding pytest coverage, improving reliability and speeding up debugging/troubleshooting across environments.
- Collaborated via open-source code reviews and discussions, aligning contributions with clean, efficient, testable code standards.

University at Buffalo

Oct 2024 - Dec 2025

Research Assistant

Buffalo, USA

Developed a multi-head Transformer for multi-label classification of cognitive-affective states using AU/VA features (DAiSEE), building end-to-end training + evaluation pipelines achieving 78.08% test accuracy.

- Improved model quality through late fusion of parallel Transformers and structured experimentation; analyzed results, iterated design decisions, and communicated findings to technical stakeholders.
- Integrated Qwen3 + LoRA + TRL SFT for parameter-efficient fine-tuning and cost-aware workflows; applied prompt engineering to produce structured multi-head predictions supporting scalable inference.

Centum T&S

Apr 2024 - Jun 2024

Software Engineer Intern

Bengaluru, India

Built real-time HMI applications for metro systems using protobuf, ZeroMQ, and WebSockets, improving application responsiveness/latency by 30% in operational workflows.

- Delivered end-user UI features (fault information, settings) with Handlebars.js + Node.js, focusing on maintainable full-stack development and strong UX.
- Collaborated cross-functionally to translate requirements into technical designs and documentation; supported debugging and issue resolution to improve system reliability.

HCLTech

Jan 2024 - Mar 2024

Software Engineer Intern

Chennai, India

Built an Online Video Platform using React.js, Node.js, and MongoDB; improved system performance by 25% and reduced authentication latency by 15%.

- Implemented secure APIs and features (JWT authentication, RBAC), upload/playback workflows, and streaming support; debugged issues end-to-end and improved production readiness.
- Deployed using Docker on GCP, enabling repeatable builds and reliable service operation in a cloud environment.

Projects

Video Affective State Analyzer (Qwen + LoRA) | <https://huggingface.co/spaces/lakki03/AU-LLM-Demo>

- Built an AI-powered application that predicts affective states from AU-driven video features and generates an LLM explanation for interpretability and user-facing insights.
- Fine-tuned Qwen using LoRA (parameter-efficient, cost-aware) across four label heads, outputting calibrated 0–3 scores with confidence distributions.
- Shipped a Gradio + Hugging Face Spaces demo with reproducible examples and clear, documentation-friendly outputs (top AUs + confidence).

Trip Planner (Multi-Agent System: CrewAI + Gemini + Search Tools)

https://github.com/lakki12233/trip_planner_crewAI

- Designed a multi-agent / agent-based architecture (city selector, local expert, concierge) delivering end-to-end user journeys through coordinating agents.
- Integrated Gemini LLMs with tool-calling agents (web search + website reader; fallback), improving robustness for real-world queries and failure scenarios.
- Implemented reliability features (provider switching, timeouts) and Markdown/PDF export to support scalable usage and troubleshooting.

Visual Question Answering (VizWiz) | https://github.com/lakki12233/VQA_bot

- Built a VQA system combining vision + language modeling for noisy real-world images; achieved 70% test accuracy in answer prediction.
- Integrated CLIP (ViT-B/32) with a lightweight classifier and evaluation workflow.

Road Lane Line Detection (OpenCV) | <https://github.com/lakki12233/Road-lane-line-detection>

- Implemented a low-latency lane detection pipeline using OpenCV + Python achieving 93.8% accuracy.

Technologies

- **Languages:** Java, Python, C++, SQL, Typescript, Bash, HTML, CSS
- **AI & ML:** LLM fine-tuning (LoRA, TRL SFT), Transformers, Deep Learning, Prompt Engineering, Model Training/Evaluation/Inference Pipelines, Experimentation & Analysis
- **Frameworks/Tools:** PyTorch, Gradio, Hugging Face Spaces, OpenCV, NumPy, Pandas, scikit-learn, Matplotlib, Langchain, Jupyter
- **Full-Stack:** React.js, Node.js, MongoDB, REST APIs, JWT Auth, Role-Based Access (RBAC)
- **Systems/Cloud:** Docker, GCP deployment, Reliability/Scalability, Performance/Latency Optimization